

Experimental data.

Determining the laminar flame speed and emission spectra in laminar Bunsen flames

May 2026

Learning objectives:

The learning objectives of this assignment are:

- to determine the laminar flame speed of premixed methane-air and hydrogen-air flames,
- to analyse flame structure using OH-PLIF and OH* chemiluminescence imaging,
- to investigate flame emission spectra under lean and rich conditions,
- to compare the combustion characteristics of methane and hydrogen flames.

You should be able to solve the experimental part of the assignment using the lecture notes, the explanations provided during the lectures, and the explanations contained in this booklet. However, you are encouraged to consult additional literature sources and reliable online references to support your analysis and discussion. Remember to cite relevant literature if you decide to use it.

If something is unclear, please contact a.silvacorreia@tudelft.nl or f.dedomenico@tudelft.nl.

1] Introduction

Laminar flame speed is a fundamental quantity used to characterise the combustion properties of a fuel-oxidiser mixture. This quantity, S_L , is a function of pressure, temperature, equivalence ratio and reactant composition. Laminar flame speed also provides a basis for understanding turbulent flame propagation, which is important for predicting flame stabilisation and the design of combustors in gas turbine engines.

In this assignment, you will analyse the laminar flame speed of premixed methane and hydrogen flames using experimental data acquired under both lean and rich conditions.

In combustion science, optical diagnostics are powerful tools for visualising flame structures and measuring scalar and vector quantities in reacting flows. In the experiment, OH planar laser-induced fluorescence (OH-PLIF) and OH* chemiluminescence were used to visualise the flame structure and the reaction zone. In addition, a spectrometer was used to record the flame emission spectra under lean and rich conditions, to identify the radicals responsible for the different flame colours.

You have been provided with OH-PLIF and OH* chemiluminescence images to identify the flame front and analyse flame behaviour. These images will allow you to calculate the laminar flame speed of premixed hydrogen and methane flames. You will use flame emission spectra to determine the species present in the flame.

2] Experiment Description

(2.1) Laminar Flame Speed

A laminar flame typically stabilises at the location where the flow velocity balances the consumption of reactants by chemical reactions together with the diffusion of heat and chemical species across the flame front. In premixed fuel-oxidiser systems, the flow velocity component perpendicular to the flame surface balances the laminar burning velocity of the mixture. This principle is used in the Bunsen burner configuration.

A laminar premixed Bunsen flame exhibits a characteristic conical shape, as shown in Fig. 1(b). The flame cone angle adjusts such that the normal component of the burner exit velocity balances the laminar burning velocity of the mixture. As illustrated in Fig. 1(a), the cone half-angle, α , can be determined from optical images.

You can calculate the unburnt gas velocity, V_u , from the mass flow rate and density of the incoming unburnt gases and the burner cross-section. The laminar burning velocity is then obtained from:

$$S_L = V_u \sin \alpha$$

In the OH-PLIF measurements acquired on the Bunsen burner, the flame front is identified by the transition between regions with negligible OH fluorescence signal and regions with strong OH fluorescence signal. This transition corresponds approximately to the reaction zone, where heat release and radical production are concentrated. In laminar premixed methane and hydrogen flames at atmospheric conditions, the flame thickness is typically of the order of a few hundred micrometres.

In the data of this assignment, the laminar Bunsen flames are imaged using OH-PLIF, a digital camera, and a UV camera equipped with a bandpass filter suitable for detecting OH* chemiluminescence. The equivalence ratio is varied while maintaining constant thermal power. You will use the acquired images to determine the laminar flame speed under these operating conditions.

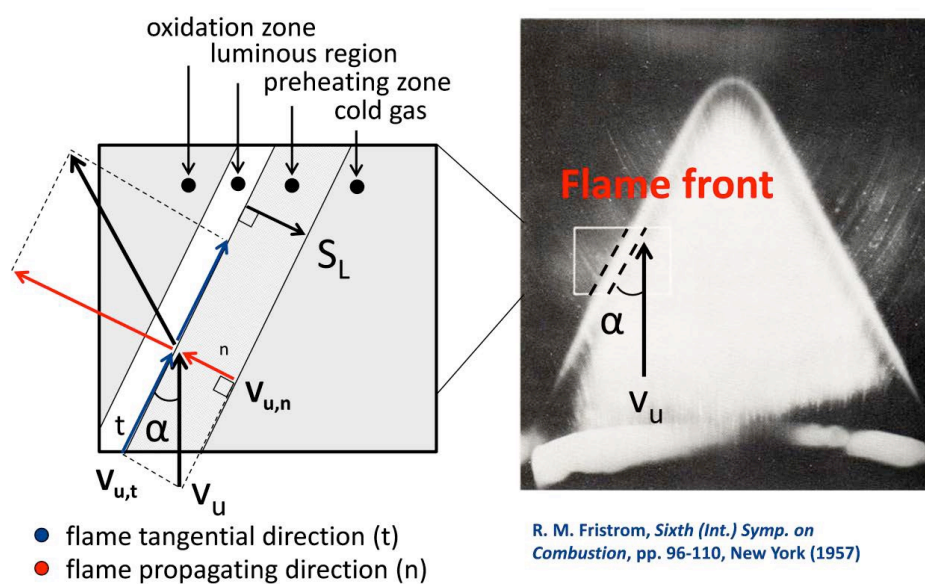


Figure 1: (left) Laminar flame speed from flow velocity components. (right) Image of a Bunsen flame

(2.2) Spectroscopic Emission Measurements

The spontaneous flame emission from methane-air Bunsen flames is measured under different operating conditions using a spectrometer. You have received the emission spectra of premixed methane-air flames.

The spectrometer collects the light emitted by electronically excited species in the flame and separates it into its constituent wavelengths. This allows the identification of specific radicals, atoms, and molecules, such as CH^* , C_2^* , CO_2^* , based on their characteristic light emission features.

By analysing the measured spectra, you can gain insight into the flame composition, combustion chemistry, and radiative processes occurring during combustion. In some cases, spectral features can also provide qualitative information about flame temperature and soot formation.

3] Laboratory Setup

3.1) Bunsen burner



Figure 2: (left) Picture of a Methane-air premixed Bunsen burner; (right) Picture of a hydrogen-air premixed Bunsen burner

The experimental setup consists of a laminar Bunsen flame burner, as shown in Fig. 2 (methane-air flame on the left and hydrogen-air flames on the right). The reactants enter the burner through a pipe with a length of approximately 50 diameters to promote complete mixing and flow homogenisation before reaching the burner exit.

For the premixed methane flames, the Bunsen burner has an inner diameter of 21 mm. For the premixed hydrogen flames, the burner inner diameter is 4 mm.

The fuel (methane or hydrogen) is supplied from the combustion laboratory gas distribution system, which consists of pressurised gas cylinders. Compressed air is supplied by the laboratory compressor system. Both air and fuel pass through pressure regulators, control valves and mass flow controllers before entering the setup. A schematic P&ID (piping and instrumentation diagram) of the Bunsen burner is shown in figure 3.

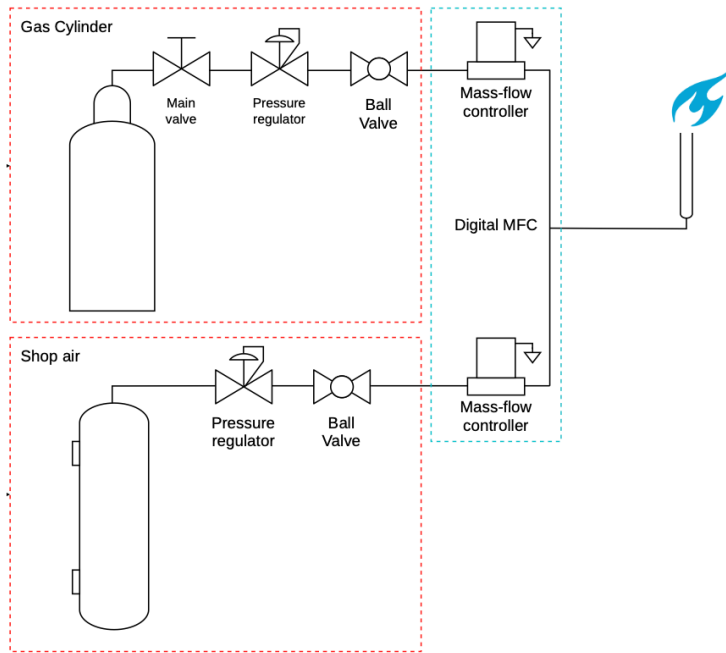


Figure 3: P&ID of the Bunsen burner experiment

Digital mass flow controllers from Bronkhorst (MFCs) are used to set the flow rates, typically expressed in NLPM (normal litres per minute) or SLPM (standard litres per minute). Although reported as volumetric flow rates, these quantities effectively correspond to mass flow rates referenced to specified temperature and pressure conditions

- **Normal conditions (ln/min):** a temperature of 0 °C and a pressure of 1013.25 hPa(a) is chosen and these reference conditions are indicated by the underlying letter 'n', in the volume unit used. The measurement method for the direct thermal mass flow is always based on these reference conditions, unless otherwise requested.
- **Standard conditions (l/min):** in this case, the reference conditions are based on 20°C instead of 0°C. Pay attention to this difference, because mixing up these reference conditions causes a 7% error.

The mass flow controllers are designed such that they supply a constant mass flow irrespective of the incoming gas temperature and pressure.

3.2) Diagnostics

A) OH-PLIF and OH* chemiluminescence

In your lectures, you have explored the principles of PLIF and chemiluminescence measurements. In the experiment, a dye laser is used as the excitation source for the OH-PLIF measurements. Unlike a conventional fixed-wavelength laser, a dye laser is tuneable, meaning that its output wavelength can be adjusted very precisely. This is essential for laser-induced fluorescence measurements, as the laser wavelength must match a specific absorption transition of the OH radical (see Figure 4). In this

experiment, the laser is tuned to a wavelength corresponding to the OH A-X (1,0) Q₁(8) absorption line at 283.636 nm.

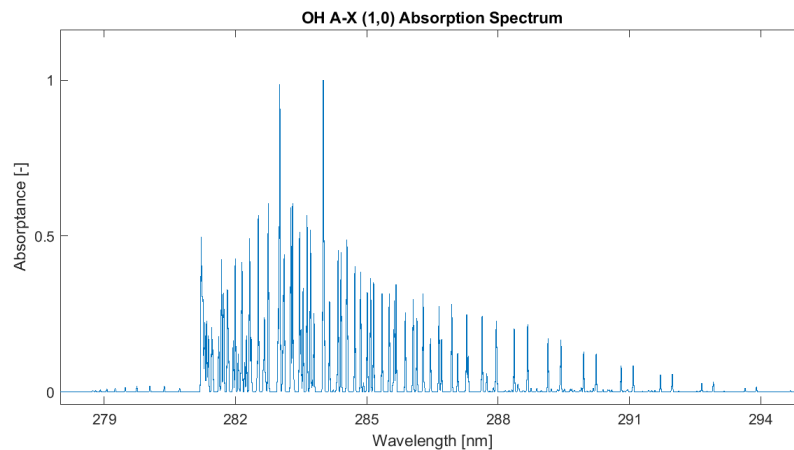


Figure 4: OH absorption spectrum within the OH A-X (1,0) transitions.

When the OH radicals absorb the laser light, they are excited to a higher electronic energy state. After a very short time, the excited molecules relax back to lower energy states and re-emit light at slightly different wavelengths. This emitted light is referred to as fluorescence, as discussed during the lectures. Since the fluorescence signal originates only from regions containing OH radicals, OH-PLIF provides images of the reaction zone of the flame.

The laser beam is shaped into a thin laser sheet using a combination of optical lenses. Typically, a collimating and cylindrical lens arrangement is used to expand the beam in one direction while maintaining a small thickness in the perpendicular direction. This creates a thin planar illumination region that intersects the flame, allowing two-dimensional imaging of the OH distribution within a selected plane of the flame.

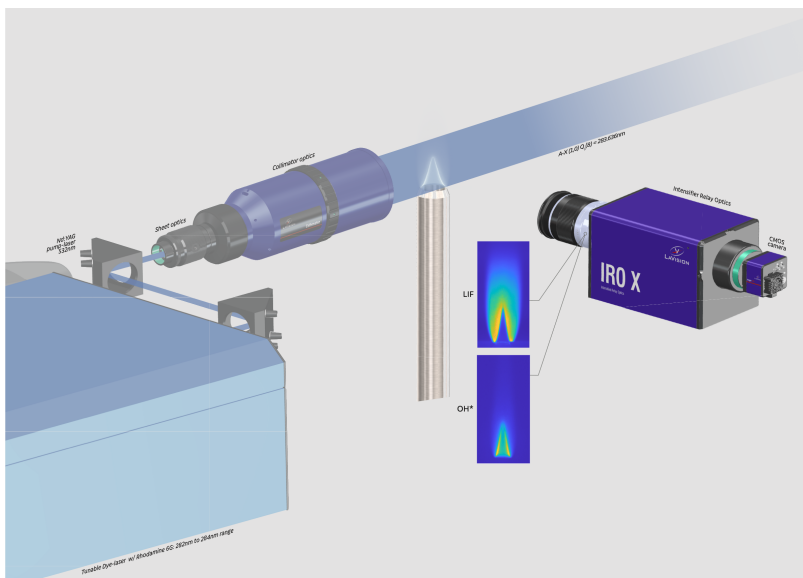


Figure 5: Schematic of the OH-PLIF measurement system

The fluorescence signal is recorded using a UV-sensitive intensified camera. A narrowband optical filter centred around the OH fluorescence wavelength (310 +/- 10 nm bandwidth) is placed in front of

the camera lens. The purpose of this filter is to reject scattered laser light at the excitation wavelength and isolate the weaker fluorescence emission from the OH radicals. Without this filtering, reflections and scattering of the laser beam would dominate the recorded signal. A schematic of the acquisition system used for the PLIF measurements is shown in figure 5.

The same intensified camera is also used to record OH* chemiluminescence images. In this case, the laser is turned off, and the camera exposure time is increased. OH* chemiluminescence originates from electronically excited OH radicals naturally produced during combustion reactions. The spontaneous emission around 308 nm provides a line-of-sight image of the heat-release region without requiring external laser excitation.

B) Spectrometer

A series of lenses and fiber optics, as shown in Fig. 6, is used to collect and focus the light emitted by the flame onto the entrance slit of a spectrometer (Thorlabs CCS100 Compact Spectrometer, spectral range 350–700 nm). The spectrometer contains a diffraction grating that spatially separates the incoming light into its constituent wavelengths. The dispersed light is then recorded by a CCD detector, enabling measurement of the flame emission spectrum with sufficient spectral resolution to distinguish important combustion radicals such as CH* and C2*.

The optical collection system increases the amount of light coupled into the spectrometer and improves the signal-to-noise ratio of the measurements. Each chemical species emits radiation at characteristic wavelengths corresponding to specific electronic or vibrational transitions. By analysing the measured spectrum, it is therefore possible to identify the excited species present in the flame.



Figure 6: Schematic picture of the optical arrangement used for acquiring the flame emission spectra.

(3.3) Mass flow calculations

Suppose that an air mass flow rate of 1 g/s is required. The mass flow controllers used in this experiment display the flow rate in normal litres per minute (NLPM), corresponding to the volumetric flow rate referenced to normal conditions (NTP).

For the air mass flow controllers used in this setup, the normal conditions are defined as:

- Temperature: 0° (273.15 K)
- Pressure: 1 atm

Under these conditions, the density of air is approximately $\rho_{air} = 1.293\text{ kg/m}^3$.

The equivalent volumetric flow rate at normal conditions can therefore be calculated from $V_n = \dot{m}/\rho_n$, where: $\dot{m}$ is the mass flow rate, and ρ_n is the density at normal conditions.

Therefore, an air mass flow rate of 1 g/s corresponds to approximately to $1\text{e-3}/1.293 \cdot 60000\text{ nlpm} = 46.4\text{ NLPM}$ under the specified normal conditions.

The temperature and pressure in the laboratory during the experiment could be taken as 300 K and 1 atm respectively.

4] Experimental procedure followed to acquire the experimental data

(4.1) Scaling and Calibration

- 1) The camera is focused to the central plane of the burner pipe.
- 2) Pictures of the setup with some mm paper for the purpose of scaling are taken.

(4.2) Start up procedure

- 1) The required fuel and air value on the air mass flow controller are set.
- 2) When the fuel flow is stable, the flame is ignited with a spark.
- 3) After successful ignition, some time is left for the flow to stabilize before acquiring the data.

(4.3) Burner operation

- 1) All the experiments are done at constant thermal power.
- 2) The OH* images and the spectrometer emission data of the flame are acquired.
- 3) Laser is turned on to acquire the OH-PLIF images.
- 4) When flashback and blowoff limits occur, the fuel is shut down immediately.

5] Preparation for the assignment: Generation of the test matrixes

Premixed flames: You will explore the flame behavior for premixed methane-air flames and premixed hydrogen-air flames

a. You should calculate the air and fuel mass flow required for the methane premixed flame at constant thermal power of **1.00 kW** for equivalence ratios (Φ) from 0.8 to 1.2 by increasing the equivalence ratio by 0.2 each step.

b. You should calculate the air and fuel mass flow required for the hydrogen premixed flame at constant thermal power of **0.250 kW** for equivalence ratios (Φ) from 0.5 to 1.3 by increasing the equivalence ratio by 0.1 each step.

H2 Premixed flame setpoints	
Φ	Stability
< 0.5	Blow-off
0.5	
0.6	
0.7	
0.8	
0.9	
1.0	
1.1	
1.2	
1.3	
>1.55	Flashback

CH4 Premixed flame setpoints	
Φ	Stability
< 0.8	Blow-off
0.8	
1.0	
1.2	
> 1.2	Flashback

Table 1: (a) H2 and (b) CH4 setpoints and stability at $P = 250W$ and $1kW$ respectively.

You should make separate tables in Excel for the experimental condition variations mentioned in 5], reporting the following quantities for the experiment conditions:

- 1) Fuel flow rate (kg/s)
- 2) Fuel flow rate (nlpm)
- 3) Air flow rate (kg/s)
- 4) Air flow rate (nlpm)
- 5) Thermal Power (kW)
- 6) Pipe flow density (kg/m³)
- 7) Pipe Reynolds number (-)
- 8) Pipe bulk velocity (m/s)

6] Data for the assignment

The provided data is organized into a main folder containing four subfolders, along with MATLAB and Python scripts for reading *.b16 files. The first subfolder, *LIF/*, contains directories for each experimental conditions following *[Fuel]_PHI[EquivalenceRatio]/* format. Each of these directories includes a *B16/* subfolder which stores the corresponding *.b16 files. The second subfolder, *OHstar/* follows the same structure. The third subfolder, *Calibration/*, contains the calibration images acquired for each fuel *[Fuel]/B16/*.b16*. These calibration images were acquired using metric paper, where each small square corresponds to 1mm. The fourth subfolder contains the images acquired from the spectrometer for premixed methane flames *[Fuel]_PHI[EquivalenceRatio]*.

- **Flame imaging:** you have received the average of 200 snapshots of OH PLIF and OH* chemiluminescence of the flames using the UV intensified camera (examples in figure 7 and 8).

- **Spectral measurements:** you have received snapshots of the emission spectra of the premixed methane flames for $\phi = 0.8, 1$, and 1.2 .

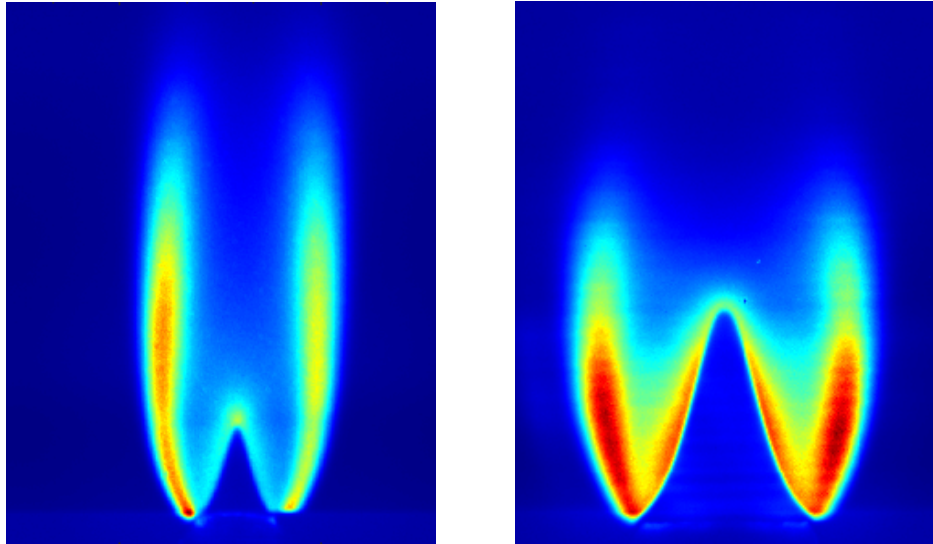


Figure 7: Acquired OH PLIF images for (left) H₂ and (right) CH₄ at $\phi = 1$ at constant thermal power of 250 W and 1 kW respectively.

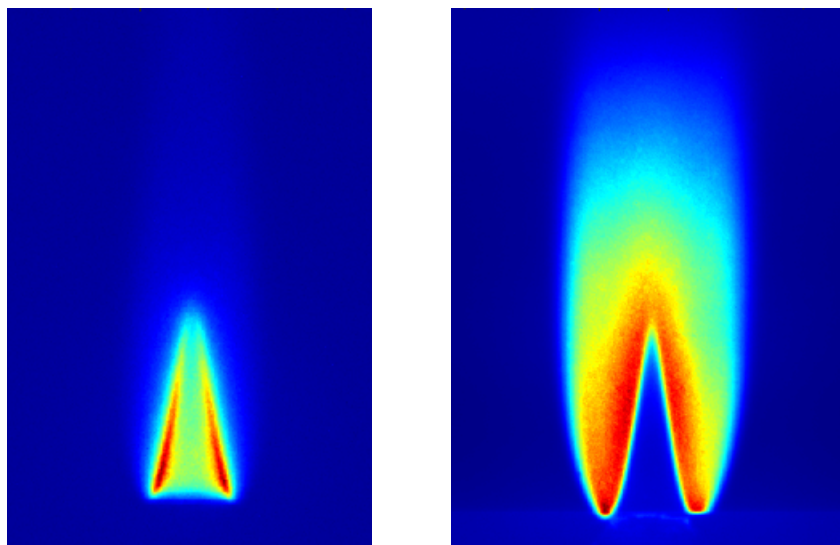


Figure 8: Acquired (left) OH* and (right) OH PLIF images for H₂ at $\phi = 0.7$ at constant thermal power of 250W respectively.

Abel inversion:

OH* is an excited combustion radical formed during chemical reactions involving the OH molecule. When OH* relaxes back to its ground state, it emits a photon, a process known as chemiluminescence. Because this emission is spontaneous and occurs throughout the flame volume, any OH* image represents a line-of-sight integration of the entire 3D flame onto a 2D plane. As a result, the recorded intensity at each pixel includes contributions from all emitting regions along the viewing direction.

This makes direct interpretation of OH* chemiluminescence difficult. For example, when plotting the intensity profile of OH* across the flame, the reactant side does not appear to go to zero because the emission from other regions along the line of sight is still captured.

However, if the flame is axisymmetric, this limitation can be overcome by mathematically reconstructing the local (radial) emission distribution from the integrated signal. This reconstruction is performed using Abel inversion, which transforms the measured line-of-sight integrated intensity into a radial emissivity distribution representing a cross-section of the flame. In other words, Abel inversion allows reconstruction of the internal radial structure of an axisymmetric flame from its projected image. Thus, Abel inversion allows reconstruction of the internal radial structure of an axisymmetric flame from its projected image.

Abel inversion relates the measured intensity $I(x)$ to the local emissivity $\varepsilon(r)$ through:

$$I(x) = 2 \int_x^{\infty} \frac{\varepsilon(r)r}{\sqrt{r^2 - x^2}} dr$$

The inversion procedure then recovers the local emissivity distribution $\varepsilon(r)$ from the measured intensity distribution $I(x)$.